

GENREG

Gradient-Free, Evolution-Based Neural Network Research

Comprehensive Project Report

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Hardware: NVIDIA RTX PRO 6000 Blackwell (102 GB VRAM)

Period: February -- March 2026

HARD CONSTRAINT: NO GRADIENTS EVER

Best Result: 85.07% CIFAR-10 (gradient-free)

27,848 evolved activation functions across 5 modalities

68.20% CIFAR-10 via evolved dendrite features (v27)

85.07% CIFAR-10 via Random Conv ELM + SPP ensemble

77.8% next-word prediction with 14 evolved parameters

47.65% via Unified Cascade with Mahalanobis features

48.08% via multi-representation pixel comparisons

Table of Contents

1. Evolved Activation Functions & Adaptive Growth
2. Evolved Activation Functions: Complete Findings
3. CIFAR-10 Dendrite Classifier (v1-v27)
4. Breaking Past 41%: Multi-Resolution Patches
5. Fold Explorer: Binary Comparisons & Truth Tables
6. Unified Cascade: Evolved Activations + Mahalanobis
7. Overnight Experiments: Maha-Only Pool Discovery
8. Motion Delta Experiments: Cortical Columns
9. Geometric Mahalanobis Cascade Architecture
10. GENREG: Random Conv ELM (85.07%)
11. PCA VAE System
12. GENREG Image Generation Training
13. CBOW Attention Model
14. Autoregressive Text Generator (Demo)
15. Semantic Layer (In Progress)

1. Evolved Activation Functions & Adaptive Neural Networks

Executive Summary

This project explores whether useful neural network components can be built entirely without gradient descent. Over four days (Feb 24-27, 2026), we evolved a catalog of 27,848 activation functions using genetic programming, then built a biologically-inspired neuron growth system that learns to organize these activations into a self-structuring network.

Key Results:

- 27,848 unique activation functions evolved across 5 modalities (MNIST, FashionMNIST, CIFAR-10, AudioSpeech, VideoMotion)
- Top activations follow the pattern $x + \sin(\text{complex_expression})$ -- identity-preserving with oscillatory perturbation
- Cross-modal transfer works: image-evolved activations perform nearly as well on audio (90.4%) as audio-evolved ones (89.7%)
- Only 50-100 activations needed to match full 27K catalog performance
- Shattering architecture achieves 38.05% on CIFAR-10, beating the L1 dendrite baseline (37.32%)
- Total network: 990K parameters, ~17 minutes training, entirely gradient-free

Activation Catalog: Galaxy Evolution

Dataset	Input Dim	Evolved	Above 20%	Best Single	Time
MNIST	784	7,219	2,669	72.1%	70s
FashionMNIST	784	8,696	3,954	75.0%	147s
EMNIST_Digits	784	7,202	3,030	75.6%	68s
CIFAR-10	3,072	9,147	1,529	28.1%	96s

After deduplication (cosine similarity > 0.999): 27,848 unique activations from 46,466 raw.

Expanded Catalog (Audio + Video)

Source	Count	Good Generalists (>55)	Great (>60)
Image-evolved	24,832	4,120	87
Audio-evolved	604	118	4
Video-evolved	626	90	1
TOTAL	26,062	4,328	92

Top Generalist Activations

9 of top 10 follow $x + \sin(\dots)$ -- identity-preserving with oscillatory perturbation:

Rank	Score	Expression
1	64.1	$x + (x + \sin((\exp(2) * \cos(x)) - 1.08))$
2	64.1	$x + (x + \sin((\cos(x)*\sin(x)) - \dots))$
3	64.0	$x + (x + \sin((\cos(2)*\sin(x)) - \dots))$
4	63.9	$x + (x + \sin((2.03*\sin(x)) - \text{neg}(\text{neg}(\dots)))$
5	63.8	$x + (x + \sin(\log(x) - 1.08))$

Cross-Modal Transfer

Activations evolved for one modality transfer remarkably well to others:

Source	MNIST	Fashion	CIFAR-10	Audio	Video
Image top-200	90.8%	82.9%	33.5%	90.4%	77.4%
Audio top-200	90.2%	82.9%	33.4%	89.7%	77.3%
Video top-200	90.4%	82.5%	32.3%	90.4%	75.3%
Mixed top-200	91.4%	82.5%	30.7%	90.8%	77.4%

Operation Enrichment in Top Performers

Operation	Top 200 Freq	Bottom 200 Freq	Enrichment
cos	0.74	0.09	8.28x
sin	0.76	0.17	4.34x
+ (addition)	1.52	0.35	4.28x
exp	0.41	0.10	3.90x
pi (constant)	0.30	0.10	3.00x

Catalog Pruning

Top-100 generalists actually OUTPERFORM the full catalog (91.9% vs 91.2%). Massive compression is possible: 50-100 activations match full 26K catalog.

L1 Dendrite Layer

The foundation: raw pixels (3072D) -> 10 basis elements with random weights -> 32 neurons with evolved activations -> 320D features -> kNN classifier.

L1 Baseline: 37.32% on CIFAR-10 (320D features). ~31,680 parameters, all randomly initialized. The only "learning" is evolutionary selection of activation functions.

Growth System: The Problem

15 growth experiments ALL performed WORSE than L1 baseline. Best growth (36.06%) was 1.35% worse. Core problem: linear terminal projections are lossy bottlenecks that destroy information.

Rank	Experiment	Accuracy	vs L1
1	child_l1_terminals	36.06%	-1.35%
2	child_l1	35.92%	-1.49%
3	higher_budget	35.84%	-1.57%
4-15	(10 experiments)	35.36%	-2.05%

Architecture Exploration & Diagnosis

Test 1 - Many Terminals (35.82%): More terminals didn't help. Problem isn't number of projections.

Test 2 - Hillock Sparsity (35.61%): Sparsity on linear transforms is still linear.

Test 3 - Simple Neurons (24.94%): Catastrophic failure. Reducing 320D to 16D destroys information.

Test 4 - Simple Sweep: Raw pixels as input (35.86%) beat L1-based input (25.33%) by ~10%.

BREAKTHROUGH: Transmutation -- per-dimension nonlinear curves. No compression. 5 neurons achieved 37.54% -- first time exceeding L1 baseline.

Shattering Architecture (38.05%)

The solution: fragment + route + reassemble -- NOT compress.

- Per-dim curves transform each feature independently (information preservation)
- Random terminal assignment (fragmentation over compression)
- Every input dimension appears in exactly one terminal
- 11 neurons, 680 terminals, 2166D features, ~17 minutes training
- Grew steadily from 37.54% (founding) to 38.05% over 7 rounds

Approach	Accuracy	vs L1	Dim
L1 Baseline	37.32%	--	320D
Original Growth (best)	36.06%	-1.35%	536D
Transmute Only (5N)	37.54%	+0.22%	1600D
SHATTERING	38.05%	+0.73%	2166D

2. Evolved Activation Functions: Complete Experimental Results

Activation Family Taxonomy

Family	Count	% Catalog	Mean Score	Description
symmetric	7,784	29.9%	37.0	Even functions, $f(-x)=f(x)$
oscillatory	6,984	26.8%	32.7	Multiple extrema, periodic
wide-range	4,502	17.3%	34.8	Large output range
tanh-like	2,296	8.8%	48.2	Saturating sigmoid (BEST)
complex	1,546	5.9%	42.7	Multi-feature
neg-monotonic	1,531	5.9%	33.6	Decreasing functions
leaky-relu	752	2.9%	31.2	Piecewise linear
bounded-flat	596	2.3%	26.4	Near-constant

What Makes a Good Activation

Good: High range, antisymmetric ($f(-x)=-f(x)$), multiple zero crossings/extrema, balanced positive/negative output (~50% negative).

Bad: Near-constant output (range < 0.5), no zero crossings, all-positive or all-negative, no antisymmetry.

Robustness to Input Noise

Activations are robust up to $\sigma=0.5$ (98.5% retention). Severe degradation only at $\sigma=2.0$.

Speech-to-Text Experiments

Best STT result: 71.2% CER with evolved 2-layer activations, ridge regression, segment decode.

Critical finding: Identity (no nonlinearity) achieves 73.6% CER. Evolved activations only improve by ~2.5 percentage points. The nonlinearity is NOT the bottleneck -- the random projection + ridge framework is.

Pyramid evolution for STT FAILED: 91.9% CER vs 71.2% baseline. Evolving per-task is worse than pre-evolved catalog.

Novel connection methods (6 tested): All used centroid classifier which proved fundamentally too weak. Best novel: Activation Fingerprint at 84.9% CER. Ridge regression with ANY activation beats centroid with BEST features.

Snake Game Evolution

Original catalog (24.0 fitness) outperformed expanded catalog (21.8). Multiple runs over 200 generations, fitness range 20.8-24.3.

3. CIFAR-10 Dendrite Classifier: 38.32% to 68.20%

Executive Summary

27 experimental versions over ~7 days. Starting from 38.32% baseline with single kNN classifier, pushed to 68.20% -- achieved entirely without backpropagation.

The system uses: L1 dendrite features (evolved random projections with 25,842 activation curves), kNN classification with distance weighting, Rich meta-features (second-order kNN confidence signals), and evolved weight fusion across multiple feature sources.

Full Accuracy Progression

Version	Accuracy	Key Innovation	Phase
v1	38.32%	Single L1 kNN, cosine distance	Foundation
v2	39.50%	Feature subgroup bagging (64 groups)	Foundation
v3	40.49%	Fisher-weighted bagging	Foundation
v4	44.73%	Multi-L1 ensemble (4 seeds)	Foundation
v7	46.77%	Multi-L1 ensemble (8 seeds)	Foundation
v10	48.54%	Rich meta: kNN->meta->2nd kNN	Meta
v11	49.47%	Evolved weight fusion	Meta
v13	50.32%	Data augmentation (H-flip) + TTA	Augment
v16	51.30%	6x augmentation + evolved fusion	Augment
v17	51.93%	Multi-feature: patches+res8+gray	Multi-feat
v20	52.80%	Scaled v17: 48 seeds, XRich-TTA	Multi-feat
v21	58.31%	8 NEW feature types +5.5%	Diversity
v25	65.18%	HOG+DCT+LBP+SPHist discovery	Discovery
v26	43.35%	Evolved fold/unfold cascade	Cascade
v27	68.20%	Ultimate: optimal per-type treatment	Ultimate

Phase 4: Feature Diversity Breakthrough (v21)

The single largest accuracy jump: +5.51% in one version. Even INDIVIDUALLY WEAK feature types (35-47%) provide orthogonal information that dramatically improves fusion.

Type	Dim	Raw kNN	Rich Meta	Information
Sobel-Gray	1024	35.43%	42.04%	Edge magnitude
Sobel-RGB	3072	39.33%	43.29%	Per-channel edges
Center16	768	39.21%	43.04%	Object focus
ColorDiff	3072	40.15%	42.80%	R-G, R-B, G-B channels
Res12	432	45.90%	46.83%	Avg-pool 12x12

Phase 5: Novel Description Discovery (v25)

Eight novel feature extraction methods. Critical discovery: L1 evolution HURTS good features.

HOG raw kNN = 56.47%, but with L1 dendrite evolution = 48.88% -- a 7.6% LOSS.

DCT drops from 53.12% to 51.40%. Haar drops from 43.64% to 34.19%.

These features are already strong; evolved random projections destroy their structure.

Phase 7: Ultimate Fusion (v27 -- 68.20%)

Feature	Processing	Seeds	Accuracy	Fusion Weight
HOG	Raw + rand-subspace	6	58.42%	3.659
DCT	Raw + rand-subspace	6	53.83%	2.388
ColorDiff	L1 evolution	6	41.72%	2.842
LBP	Raw + rand-subspace	4	43.29%	2.138
6xAug-kNN	Augmented raw	1	44.56%	2.403
SPHist	L1 evolution	4	42.19%	1.659
SobelRGB	L1 evolution	6	38.67%	1.053
Patches	L1 evolution	4	40.20%	0.479

The Four Orthogonal Axes of Image Description

Axis 1 - Shape (HOG): Gradient orientation histograms. Evolved weight 3.659 (highest).

Axis 2 - Frequency (DCT): Frequency band decomposition. Weight 2.388.

Axis 3 - Color (ColorDiff): Color relationships (R-G, R-B, G-B). Weight 2.842.

Axis 4 - Texture (LBP): Local binary patterns. Weight 2.138.

Error correlation confirms orthogonality: when HOG errs, only 58-61% of the time does DCT also err. Each axis recovers ~40% of errors from any other.

Key Principles

1. Feature DIVERSITY >> Feature QUANTITY
2. Match processing to feature nature (L1 hurts HOG/DCT, helps ColorDiff/Sobel)
3. Rich meta-features are essential (+3-5% absolute)
4. Evolved fusion outperforms equal weighting (68.20% vs ~60%)
5. Compression needs good features first
6. Augmentation remains fundamental

4. Breaking Past 41%: Multi-Resolution Patches

Experiment Results

Exp	Config	Founding	Final	Neurons	Time
C (BEST)	3 Global + 9 Multi-Res	41.48%	42.59%	22	85min
E	5 Global + 6 Grid + Deep	41.34%	41.79%	15	102min
B	5 Global + 6 Grid	40.67%	41.72%	19	37min
D	3 Global + 3 Free + 3 Raw	41.47%	41.60%	11	24min

Exp C beat raw pixel kNN baseline (42.45%) with 42.59%. First time architecture exceeded simple kNN on raw pixels.

Key findings:

- Founding quality determines final accuracy (near-perfect correlation)
- Growth shows diminishing returns: first 3 rounds +0.90%, last 7 rounds +0.21%
- Multi-resolution captures info at different spatial granularities (fine=texture, medium=parts, full=context)

5. Fold Explorer: Binary Comparisons & Truth Tables

The Headline: pixel_compare Beats ALL Weighted Folds

Just 'is pixel A > pixel B?' for 8192 random pairs. ZERO learned weights. 38.48% accuracy, beating all weighted folds.

Rank	Fold	Best	Active %	vs Raw
1	pixel_compare	38.48%	50.0%	+1.56%
2	binary_sparse	37.39%	50.0%	+0.47%
3	ternary	37.35%	~33%	+0.43%
4	competitive_wta	37.20%	12.5%	+0.28%
9	ring	10.41%	~0%	-26.51%

Stacked Binary Fold

Peak at L4 (39.48%). Active fraction collapses: 50% -> 25% -> 19% -> 15% -> 13%. Binary-on-binary loses info.

Transmutation: L3 Truth Tables = 40.91%

v2: 16384 binary L2 -> 24576 3-input truth tables at L3 = 40.91% (best ever +3.99% vs raw). 193 unique truth tables out of 384 groups.

Multi-Representation pixel_compare = 48.08%

9 representations (raw, sobel, gradient, blur, local contrast, coarse, color diff) pooled. Evolution draws ~10% from each.

L1 (16384 pc): 47.87%, L1+L2 (32768): 48.08%. The representation matters more than depth.

6. Unified Cascade: Evolved Activations + Mahalanobis

Architecture

Layer-by-layer gradient-free feature learning. Each layer evolves neurons that combine pairs of input features via 12 binary operations, then pass through learned 200-point activation curves. Mahalanobis distance features appended. All layers accumulated into growing pool. kNN on full pool.

Per-neuron: output = activation_curve(op(X[idx_a], X[idx_b]))
Operations: ADD, SUB, MUL, SAFE_DIV, MEAN, MIN, MAX, ABS_DIFF, CLAMP, GT, LT, THRESH_A

Experiment Results

Config	Baseline	Best Pool kNN	Best Layer	Improvement
16K neurons, 3 layers	21.51%	29.10%	L2	+7.59%
2K neurons, 12 layers	21.51%	25.81%	L4	+4.30%
Augmented, small, 19L	22.87%	16.91%	L2	-5.96%

Key Findings

What works: Pool accumulation prevents layer destruction. kNN fitness > NCC fitness. Gene locking saves 70-95% of generation budget.

What fails: Layers plateau after L2-L3. Fitness metric mismatch (62-68% train, 13-14% test). Compression hurts. Subsampling overfits.

7. Overnight Experiments: Maha-Only Pool Discovery

ALL-TIME BEST: 47.65% kNN (Maha-Only Pool)

Rank	Config	Best kNN	Pool Type
1	D6: pop 60, 1024n, augment	47.65%	Maha-only L1
2	D2: pop 100, 1024n	46.92%	Maha-only L1
3	D4: pop 60, seed=123	46.68%	Maha-only L1
4	C6: pop 60, 1024n	39.34%	All-features L3
5	B3: pop 60, eval=4K/6K	38.73%	All-features L4

Critical Finding: Maha-Only >> All-Features

Maha-only pool (10D per layer) dramatically outperforms combined projection+maha features (1034D per layer) for kNN:

- Maha-only L1: 46-48% kNN on 10D
- All-features best: 38-39% kNN on 3000-4000D

Why: kNN suffers curse of dimensionality. 10D >> 1034D for kNN. Maha features are class-distance representations, normalized and comparable. Projection features are noisy.

8. Motion Delta Experiments: Cortical Columns

The Cortical Column Discovery

128 neurons x 40 layers: 28.87% (L1) -> 37.73% (L10) -> 40.25% (L20) -> 43.51% (L40). Still climbing.

Exp	Config	Peak kNN	Peak Layer
Base	no delta, 1024n	28.41%	L8
1	delta+raw, 1024n	37.60%	L5
5	delta+raw, 256n, 20L	39.42%	L20
7	delta+raw, 256n, 40L	42.34%	L40
8 (BEST)	delta+raw, 128n, 40L	43.51%	L40

Key Findings

1. Delta validated: +15% absolute (43.51% vs 28.41%)
2. Cortical column principle: many thin layers >> few thick layers (128n x 40L > 1024n x 10L)
3. Pool growth rate is the bottleneck: slow growth = healthy classifiers
4. Ridge as EVAL metric peaks at 52% (L20) -- best Ridge ever
5. Delta + feed-raw is winning combo: raw provides foundation, delta provides trajectory

9. Geometric Mahalanobis Cascade (Proposed Architecture)

Hypothesis: Each system plateaus because it exhausts the information in its fixed frame. Mahalanobis plateaus at 54% (one projection). Evolution plateaus (blind to class structure). Interleaving may break both.

Architecture: Evolve projection -> Measure Mahalanobis geometry -> Normalize -> Evolve next projection on normalized space -> Repeat.

Each layer sees a DIFFERENT landscape because previous Mahalanobis transform rotated/scaled space. The cascade is: project -> measure shape -> project relative to shape -> measure new shape.

Status: Proposed, awaiting experimental validation. Target: exceed 68.20% (v27).

10. GENREG: Random Conv ELM (85.07% -- Current Best)

Architecture

Random fixed convolutional filters + analytical ridge regression. No learned parameters in feature extractor. No gradients anywhere.

Pipeline: Random conv filters (multi-scale 1x1, 3x3, 5x5) -> Mixed nonlinearities (ReLU, abs, leaky) -> Spatial Pyramid Pooling (1x1, 2x2, 4x4) -> Ridge regression $W = (H'H + \alpha I)^{-1} H'Y$

Key: 2048 filters per layer, 3 layers, ~12,294 features per seed.

Accuracy Progression

Method	Accuracy	Notes
Single RGB seed	69.92%	Baseline ELM
10 RGB seeds	75.02%	Seed diversity
7 representations	77.95%	Repr diversity >> seed
Multi-repr + aug (28 clf)	80.19%	Greedy ensemble
SPP + multi-repr (11 clf)	84.03%	SPP breakthrough (+5-6%)
SPP + f2048 + diversity	85.07%	13 clf, BEST RESULT
Equal avg ensemble	85.49%	Simple averaging

10 Representations Tested

Repr	Single Seed Acc
ColorEdges (Sobel per RGB)	71.44%
LocalContrast	70.78%
MultiRes (blur+residual)	70.54%
RGB (original)	69.92%
LogColor	69.25%
OpponentColor (L, R-G, B-Y)	67.19%
SobelEdges	66.23%
Grayscale	65.09%
GradOrient	62.60%

Post-Processing Attempts (All Failed)

- TTA on ensemble: 85.26% (hurts -- redundant with diversity)
- Stacked ELMs: 85.49% (neutral)
- CMA-ES weights: 81.69% (much worse)
- Maha cascade: 83.41% (below baseline)
- Nonlinear expansion (3202d): 84.99% (helps but < averaging)

Diagnosis: Training accuracy at 99.8% means CMA-ES has no discriminative signal. The 14.4% generalization gap is in features, not the classifier.

Lessons

1. Diversity beats depth: 10 representations > 100 same-repr seeds
2. Analytical > Evolved for classifiers: Ridge in closed form beats any evolved classifier
3. SPP is disproportionately valuable: +5-6% from spatial structure alone
4. Compression destroys; expansion helps (sometimes)
5. Post-processing saturated logits is futile

11. PCA VAE System

Linear VAE via PCA for image generation pipeline. Pure linear algebra, no gradients.

Encoder: flatten(x) -> center -> project via top 2048 PCA eigenvectors

Decoder: latent @ components + mean -> reshape -> clip to [0,1]

Parameter	Value	Description
Input Resolution	96 x 96 x 3	RGB images
Pixel Dimension	27,648	Flattened input
Latent Dimension	2,048	PCA components kept
Variance Captured	96.4%	Information preserved
Training Images	105,000	STL-10 labeled + unlabeled

Integration: CLIP encodes text -> GENREG maps conditioning to latent (2048D) -> PCA VAE decodes to image.

Important: Latent space is NOT Gaussian. Random $N(0,1)$ sampling produces poor results. Manifold navigation required.

12. GENREG Image Generation Training Log

Run	Config	Result	Key Finding
1	pop=20, hidden=128, CLIP sim	MSE 5.4-8.7	No selection pressure
2	pop=50, hidden=256, norm MSE	MSE 4.35-5.5	Getting worse
3	pop=50, W2=0, no proteins	MSE 1.12->8.92	No elitism = drift
4a	Direct latent, flat scale	39.7% improvement	Evolution works
4b	Direct latent, PER-DIM scale	90.2% improvement	KEY INSIGHT

CRITICAL DISCOVERY: CLIP->PCA mapping is impossible. Linear $R^2 = 0.003$. CLIP = semantics (what), PCA = pixels (how) -- orthogonal spaces.

KEY INSIGHT: Per-dim mutation scaling (proportional to z_{train} std per dim) is critical for high-dimensional evolution. Run 4b achieved 90.2% improvement vs 39.7% with flat scaling.

13. CBOW Attention Model

Version	Architecture	Params	H@1	Innovation
v36-A	N-gram weighted sum	27	77.3%	5-gram types baseline
v44-A	Self-attention + frozen n-gram	35	77.6%	True self-attention
v45-D	Re-evolved n-gram	27	77.7%	Better ES optimization
v47-B/v48-A	Context-adaptive n-gram	54	77.8%	BEST: per-signal ctx weights

Key findings:

1. N-gram tables dominate for CBOW -- they encode $P(\text{word}|\text{context})$ directly
2. Embedding-based attention CANNOT compete (0.2-1.2% H@1 -- useless)
3. Context-adaptive signal weighting provides genuine small improvement
4. ES optimization effective for up to ~80 parameters
5. Ceiling around 78% H@1 for this task/corpus

14. Autoregressive Text Generator

14 trainable parameters. No backpropagation. Weighted ensemble of n-gram lookup tables with evolutionarily-learned weights.

Performance: 60.7% Hits@1, 91.0% Hits@5, Mean Rank 2.8. Training: 71 seconds.

The 4-gram gets weight 5.7 (highest), bigrams 0.2-1.2. Two PMI signals have NEGATIVE weights (hurt prediction). Negative frequency weight acts as anti-frequency bias.

CPU is 3x faster than GPU (hash table lookups, not matrix math): 5,524 tok/s CPU vs 1,229 tok/s GPU.

Limitation: degenerates after 5-10 tokens into function word loops. 4-word context window, no long-range memory.

15. Semantic Layer (In Progress)

NEW DIRECTION: Moving beyond statistical co-occurrence to actual semantic understanding.

Architecture: Frozen tokenizer v4 embeddings (5000 x 128D) -> Evolved semantic projection (128D -> 64D) -> Predict next word in semantic space via cosine similarity. 8,260 evolved parameters. No lookup tables.

Fitness function (explicitly targets semantics):

- Coherence (35%): correct word scored > frequency-matched distractor
- Rare-word accuracy (30%): prediction on low-frequency words (n-gram's blind spot)
- Anti-ngram bonus (20%): correct specifically where n-gram fails
- Analogy completion (15%): king - man + woman = queen geometric reasoning

Fully GPU-vectorized: entire population evaluated in single batched forward pass. Zero Python loops in eval path.

Status: Architecture built, initial evolution showing coherence climbing from 0.54 (random) to 0.62+ within 30 generations. Awaiting GPU reset for full training run.

16. Comprehensive Failure Log

Feature Engineering Failures (v18)

- Larger L1 (48 neurons): diminishing returns
- Gradient boosting on kNN residuals: residuals not smooth enough
- Mixup augmentation: confuses kNN distance
- Crop TTA: CIFAR-10 objects too small
- Per-dimension weighting: hurts cosine similarity

Classification Strategy Failures (v22)

- ECOC: 16.43%/15.85%/9.22% -- COMPLETE FAILURE
- Temperature-scaled softmax: marginal (50.33-50.84%)
- Multi-k evolved sources: 46.56% (worse than fixed k=7)
- Feature interaction products: all worse

Architecture Failures

- Linear terminal projections: lossy bottleneck (all 15 growth experiments failed)
- Deep fold cascade: destroys information (38.79%)
- Double evolution (L1 on compressed): counterproductive
- Hierarchical classification: never beat flat
- Lottery Ticket Evolution: 23.64% (search space too large)
- OpenAI-style ES on ConvNets: ~27% (too many parameters)
- Neural Cellular Automata: 47.69% (too few parameters)

Post-Processing Failures (on 85.49%)

- TTA: 85.26% (redundant with diversity)
- Stacked ELMS: 85.49% (neutral)
- CMA-ES weights: 81.69% (much worse)
- All Maha cascade versions: never beat baseline
- Random compression in cascade: destroys signal irreversibly